

Least Squares Fit: $\hat{y} = m\hat{x} + b$

Given the following three points, find the line that fits them: $(-4, -2), (2, -8), (3, -7)$.

Mean:

$\bar{x} = \frac{1}{n} \sum x_i$	
$\bar{y} = \frac{1}{n} \sum y_i$	

Deviations:

$x_i - \bar{x}$	$y_i - \bar{y}$

Sum of Squares:

$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
$\sum (x_i - \bar{x})^2 =$	$\sum (y_i - \bar{y})^2 =$

Sum of Cross Deviations: $(x_i - \bar{x})(y_i - \bar{y})$

$\sum (x_i - \bar{x})(y_i - \bar{y}) =$

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Slope:

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} =$$

Intercept:

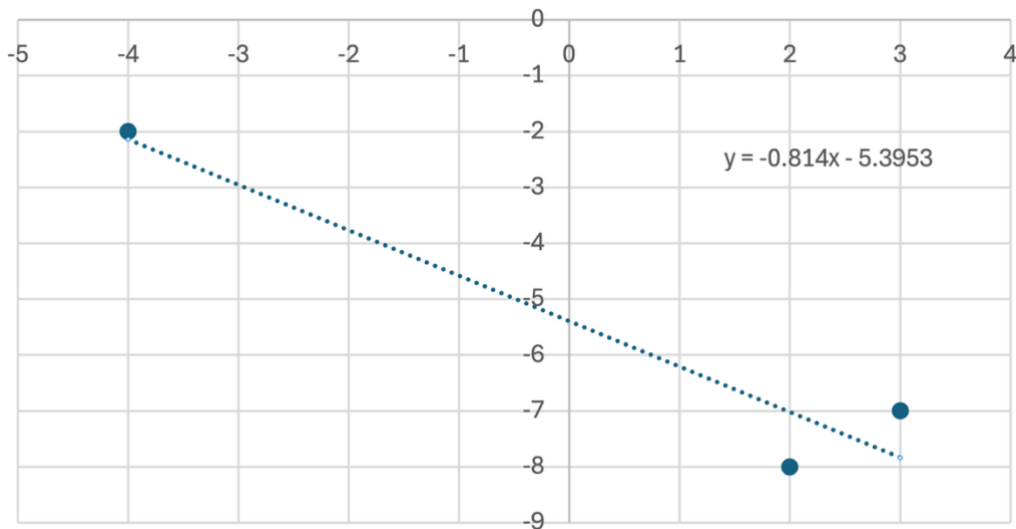
$$b = \bar{y} - m\bar{x} =$$

Final Line:

$$\hat{y} = m\hat{x} + b =$$

Predictions:

x_i	y_i	$\hat{y}_i = m\hat{x}_i + b$	$(\hat{x}_i, \hat{y}_i)$	Residuals: $e_i = y_i - \hat{y}_i$
-4	-2			
2	-8			
3	-7			



Mean Square Error:

$$\frac{1}{n} \sum (y_i - \hat{y}_i)^2 =$$

Interpretation: